Fiscal Year:	FY 2009	**Task Last Updated:**	FY 07/07/2009
PI Name:	Pyle, Barry H Ph.D.		
Project Title:	Bacterial Physiology and Virulence on Earth and in Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	59717	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-04
Start Date:	09/01/1999	**End Date:**	01/31/2009
No. of Post Docs:	4	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	10
No. of Bachelor's Candidates:	10	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Foton/Shuttle		
Flight Assignment:	Foton (changed from Shuttle in late 2005; info from ARC 9/07) Foton M2 Foton M3 STS-123 NOTE: Received extension to 1/31/2009 per A. Chu/ARC (6/2008) NOTE: Received NCE through 6/30/2008, per A. Chu (6/07) NOTE: Received NCE through 9/30/2007, per A. Chu (9/06)		
Key Personnel Changes/Previous PI:	John T. Lisle, Post Doctoral Assistant, 1999-2000 ; Margaret A. Juergensmeyer, Post Doctoral Assistant, 2000-2003 ; Elinor d. Pulcini, Post Doctoral Assistant, 2001-2004 ; Tresa. L. Goins, Assistant Research Professor, 2005-2008 ; Susan C. Broadaway, Research Assistant, 1999-2009.		
COI Name (Institution):			
Grant/Contract No.:	NCC2-1143		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	This project was planned around the objective of determining the effects of spaceflight and microgravity on bacterial growth, physiology, and virulence. Techniques used included biochemical and molecular techniques, virulence assays and physiological assays. Pseudomonas aeruginosa is a ubiquitous opportunistic bacterial pathogen that causes infections in people with weakened immunity. It has been found in spacecraft water systems. Spaceflight crews may have secondary immunodeficiency, making them more susceptible to these infections particularly on long-duration missions like those proposed to the Moon and Mars, and extended International Space Station expeditions. In spaceflight conditions, bacteria may grow faster and reach higher population numbers, as well as becoming resistant to some antibiotics. Earth experiments with modeled microgravity suggest that bacterial virulence may also be enhanced. The combination of these factors leads to a greater potential health risk to spaceflight crew members. The research provides evidence on whether or not spaceflight affects the physiology and virulence of this particular pathogen. A strain of Pseudomonas aeruginosa that produces exotoxin A was cultured on orbit and on earth to determine microgravity effects. Techniques that had been used on previous flight experiments such as plate counting, total direct microscopic counts, and metabolic fluorochromes were applied to assess growth and physiology. In addition, microscopic techniques including image analysis were employed to observe morphology, staining, and total cell count. Following the Columbia Shuttle disaster, efforts were made to re-fly the experiment on a future Shuttle mission which led to the Microbial Drug Resistance and Virulence (MDRV) experiment on STS-123 in March 2008. Additional experiments on genetic stability in bacteria were flown on Russian satellite missions Foton-M2 with in May/June 2005 with Streptomyces lividans, and Foton-M3 in September/October 2007 with S. lividans, Streptomyces coelicolor, Pseudomonas aeruginosa, Escherichia coli, Bacillus pumilus, Lactobacillus casei. The Foton M-3 mission provided an opportunity to repeat and enhance the experiment flown on Foton M-2 and expand knowledge of how living systems respond to the unique environment of space, i.e. microgravity and space radiation. These experiments have provided information on both chromosomal and plasmid gene expression as well as genetic stability. Streptomyces spp. may serve as a sensitive indicator of the effects of spaceflight on microorganisms due to their inherent genetic instability: the effects of longduration spaceflight missions may be inferred from the fundamental knowledge obtained during relatively short-term spaceflight exposure of Streptomyces specimens during the Foton M-3 flight. Detection of genetic changes in Streptomyces exposed to spaceflight factors, e.g. radiation and microgravity, would suggest that similar effects may occur in other microorganisms, especially with prolonged exposure to the same factors in long duration spaceflight. Genetic alterations (mutations) may result in enhanced virulence of pathogenic bacteria. They may also lead to changes in treatment approaches for infections, e.g. the microbes may become more or less susceptible to some antibiotics.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Changes in physiology and virulence of opportunistic pathogens such as Pseudomonas aeruginosa, combined with the well-known weakened immunity of spacecraft crew members, may affect both the risk of infections especially with prolonged exposure to factors including radiation, microgravity and individual stress. Detection of genetic changes in Streptomyces and other bacteria exposed to spaceflight factors, e.g. radiation and microgravity, would suggest that similar effects may occur in other microorganisms, especially with prolonged exposure to the same factors in long duration spaceflight. Genetic alterations (mutations) may result in enhanced virulence of pathogenic bacteria. They may also lead to changes in treatment approaches for infections, e.g. the microbes may become more or less susceptible to some antibiotics.
Task Progress:	Experiment BACTER was flown on STS-107 in January 2002, ending with the tragic loss of Shuttle Columbia. Results from the Biopack ground cultures of Pseudomonas aeruginosa strain PA103 (ATTC 29260) showed that modified incubation conditions and times, which were necessitated by in-flight incubation anomalies, had been appropriate. Growth was obtained at mid-deck ambient temperatures well into the logarithmic phase within 24 h and to the stationary phase by 51 h. It was demonstrated that P. aeruginosa can be cultured, resuspended in sterile water, and remain viable for at least two weeks at refrigeration temperature. Cell populations, respiratory activity, and production of Exotoxin A (ETA) were all consistent between experiment containers. In the MDRV Pseudomonas aeruginosa experiment, which was based on the STS-107 experiment, two strains were incubated in two media: PA01 in Lennox broth and both PA01 and PA103 in MSDM2 broth for ETA production. No differences were detected in Exotoxin A (ETA) production for either strain (PA01 or PA103) from flight vs ground incubated cultures, although some differences were demonstrated in modeled microgravity (MMG) experiments. RNA gene microarray analyses showed that, overall, 348 genes were either up- or down-regulated. Of the three variables (strain, medium, and spaceflight), the medium had the most effect followed by the strain and spaceflight. For the Foton experiments, an increase in the proportion of Streptomyces lividans strain 66 clones that retained and expessed plasmid pIJ702 genes for thiostrepton resistance and melanin production in spaceflight cultures suggested that there was selection of melanin-producing clones. This may be related to the protective effects of melanin in response to spaceflight stress factors such as radiation and microgravity. With Pseudomonas aeruginosa, Bacillus pumilis, Escherichia coli, and Lactobacillus casei DNA was isolated from flight and ground control cultures, analyzed by polymerase chain reaction (PCR) amplification, and the putative mutations identified by denaturing gradient gel electrophoresis (DGGE). These analyses detected revealed putative mutants in the E. coli flight and control cultures compared with stock of the same strain, but there were no differences between flight and ground samples in three amplicons. While PCR products from different species amplified by the same primer pair migrated similarly on an agarose gel, they migrate at different rates in DGGE because of their differing guanine/cytosine ratios, but no evidence of mutations was detected between any of the flight and ground culture pairs examined. Arginine and proline prototrophs from flight and ground cultures of S. coelicolor plated on selective media yielded a variety of pigmented isolates. PCR amplification of DNA from the two parental strains SC-236 and SC-3110 demonstrated the detection of Arginine, phenylalanine and proline gene sequences. Amplification of genes for arginine, phenylalanine and proline synthesis was demonstrated in the parent strains, but difficulties with amplifying these genes in revertants prevented further analysis. Based on the Foton data, there is no evidence of spaceflight-related genetic changes in the bacteria studied. This may be attributed to the relatively low radiation dose received by the microorganisms during the Foton flights. Mutation rates, indicated by plasmid instability, for S. lividans flight and ground samples were similar and reflect the inherent genetic

	variability of this species.
Bibliography Type:	Description: (Last Updated: 05/02/2017)
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